Conformal Replay for Off-Policy Multi-Agent Reinforcement Learning

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ABSTRACT

In off-policy multi-agent reinforcement learning (MARL), each agent perceives the other learners as part of its environment; as their policies drift, the perceived dynamics become non-stationary and previously collected transitions turn misleading. The usual remedy—shrinking the replay buffer—discards the sample efficiency that motivates off-policy learning in the first place. We recast the *staleness* of a stored transition as a question of *conformity* to the other agents' *current* policies, and answer it with conformal prediction. We define a non-conformity score that measures how surprising a transition is under the agents' current policies—or, when those policies are unavailable, under a model of the currently perceived dynamics—calibrate it on fresh transitions, and obtain distribution-free p-values. These p-values serve two purposes: (i) a large-batch-then-subsample *selection* rule that biases learning towards transitions compatible with the present, generalising importance-ratio prioritisation to settings where the ratio is unavailable or numerically unstable; and (ii) a conformal *test martingale* that detects, with an anytime false-alarm guarantee via Ville's inequality, when the perceived environment has drifted enough to invalidate the buffer. We formalise the perceived best-response value that these schemes target, clarify the (self-normalised, deliberately biased) nature of ratio prioritisation, and study the approach on two controlled cooperative environments designed to isolate perceived non-stationarity.

CCS CONCEPTS

• **Computing methodologies** → **Multi-agent reinforcement learning**; **Sequential decision making**; • **Mathematics of computing** → *Hypothesis testing and confidence interval computation.*

KEYWORDS

Multi-agent reinforcement learning; Off-policy learning; Experience replay; Non-stationarity; Conformal prediction; Sequential change detection

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1 INTRODUCTION

Many domains—robotics, logistics, healthcare, finance, and strategy games—could benefit from agents that learn to act together. Yet, in spite of impressive results on benchmarks such as the StarCraft Multi-Agent Challenge [11], multi-agent reinforcement learning (MARL) remains hard to deploy because of slow and unstable convergence [7].

A large part of this difficulty has a single, well-identified cause. Single-agent reinforcement learning assumes a *stationary* environment. In MARL, from the point of view of one agent, the other agents are part of the environment; as they update their policies, the dynamics that the agent perceives *change over time*. This *perceived non-stationarity* [7] is particularly damaging for *off-policy*, value-based methods, which owe their sample efficiency to a *replay buffer*: they repeatedly learn from transitions collected long ago, under *outdated* policies of the other agents. A common workaround is to keep the buffer small, so that stale transitions are quickly forgotten—but this throws away exactly the sample efficiency and the offline-learning capability that motivate off-policy learning.

The key observation of this paper is that *old does not imply useless*. A stored transition is harmful only insofar as the joint action taken by the other agents in it is *implausible under their current policies*. If their stored behaviour is still compatible with what they would do now, the transition is as informative as a fresh one. The problem of reusing the past therefore reduces to a problem of *selecting transitions that conform to the present.*

Prior work makes this concrete through an importance-sampling ratio between the current and past probabilities of the other agents' actions [3]. We revisit that idea, fix its mathematical formulation, and show that the ratio is often either *unavailable* (partial observability, continuous actions, parametric exploration, black-box teammates) or *numerically unstable* (its variance explodes). We then replace the explicit ratio by a *calibrated, distribution-free measure of conformity* drawn from conformal prediction [1, 20].

Contributions.

(1) We formalise the value that off-policy MARL implicitly targets as a *best-response value* against the other agents' current policies, and make explicit the quasi-stationary approximation it relies on (Section 3, Appendix B).
(2) We give a corrected account of importance-ratio reweighting: the change-of-measure identity and its support condition, the heavy-tailed instability of direct loss correction, and the fact that ratio *prioritisation* is a *self-normalised*, deliberately biased estimator rather than an unbiased one (Section 3).
(3) We introduce *conformal replay*: a non-conformity score for stored transitions, calibrated into p-values, used both to *select*

compatible transitions (Section 4.3) and to *detect* drift of the perceived environment through a conformal test martingale with an anytime false-alarm guarantee (Section 4.4).

(4) We describe two controlled cooperative environments that isolate perceived non-stationarity, and report the motivating evidence together with an evaluation protocol (Section 5, Appendix C).

2 BACKGROUND AND NOTATION

2.1 Reinforcement learning

A Markov Decision Process (MDP) is a tuple (S, A, P, R, γ) with state set S, action set A, transition kernel $P(\cdot \mid s, a) \in \Delta(S)$, reward distribution $R(\cdot \mid s, a)$, and discount $\gamma \in [0, 1)$. A policy $\pi : S \to \Delta(A)$ induces a return $\sum_{t\geq 0} \gamma^t R_t$, and the action-value function $Q_\pi(s, a) = \mathbb{E}_\pi[\sum_{t\geq 0} \gamma^t R_t \mid S_0 = s, A_0 = a]$. The optimal value $Q^* = \max_\pi Q_\pi$ is the unique fixed point of the optimal Bellman operator

$$(\mathcal{B}^* Q)(s, a) = \mathbb{E}_{s'\sim P(\cdot|s,a)}\Big[R(s, a, s') + \gamma \max_{a'} Q(s', a')\Big], \quad (1)$$

which is a γ-contraction for $\|\cdot\|_\infty$ (Appendix A). Note that the right-hand side uses Q, not Q^*: $\mathcal{B}^*$ maps an arbitrary Q to a new function, and iterating it converges to Q^* by Banach's theorem.

Deep Q-learning. DQN [8] parameterises Q_ϕ by a neural network, stores transitions in a replay buffer $\mathcal{D}$, and minimises, over uniformly sampled batches,

$$\mathcal{L}(\phi) = \mathbb{E}_{(s,a,r,s')\sim U(\mathcal{D})}\Big[\big(Q_\phi(s, a) - y\big)^2\Big], \quad y = r + \gamma \max_{a'} Q_{\bar\phi}(s', a'), \quad (2)$$

with $\bar\phi$ a periodically updated target network. A stochastic gradient step on a batch of size B reads

$$\phi \leftarrow \phi - \alpha \nabla_\phi \frac{1}{B} \sum_{i=1}^{B} \big(y_i - Q_\phi(s_i, a_i)\big)^2. \quad (3)$$

Exploration typically uses an ϵ-greedy policy: act greedily w.p. $1-\epsilon$ and uniformly at random w.p. ϵ.

2.2 Multi-agent reinforcement learning

A cooperative *stochastic game* is a tuple $(N, S, (\mathbb{U}_a)_{a=1}^N, \mathcal{T}, (r_a)_{a=1}^N, \gamma)$: N agents share the state $s \in S$, agent a picks $u_a \in \mathbb{U}_a$, the joint action $\mathbf{u} = (u_a, \mathbf{u}_{-a})$ drives the transition $\mathcal{T}(\cdot \mid s, \mathbf{u})$ and the reward $r_a(s, \mathbf{u})$. We write a for the agent under consideration and $-a$ for the others, so that $u_a \sim \pi_a(\cdot \mid s)$ and $\mathbf{u}_{-a} \sim \pi_{-a}(\cdot \mid s)$.

Independent Q-learning (IQL).. In IQL [15] each agent runs its own DQN, treating the others as part of the environment. Its loss is

$$\mathcal{L}_a(\phi_a) = \mathbb{E}_{(s,\mathbf{u},r,s')\sim U(\mathcal{D})}\Big[\big(Q_{\phi_a}(s, u_a) - y_a\big)^2\Big], \quad (4)$$

$$y_a = r_a + \gamma \max_{u'_a} Q_{\bar\phi_a}(s', u'_a).$$

IQL is simple and, despite its non-stationarity, underlies competitive value-decomposition methods such as VDN [13] and QMIX [10]; the mechanisms we develop apply to those methods unchanged, since each agent still maintains an individual Q-function. We adopt the fully observable, simultaneous-action setting for clarity; the partially observable case is discussed in Section 7.

3 THE REPLAY MISMATCH IN MARL

3.1 The perceived MDP and the best-response value

Fix agent a and freeze the others at a joint policy π_{-a}. From a's viewpoint the game collapses to an MDP $\mathcal{M}_a(\pi_{-a})$ with *perceived* transition and reward kernels

$$P^{\pi_{-a}}(s' \mid s, u_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi_{-a}(\cdot|s)}\big[\mathcal{T}(s' \mid s, u_a, \mathbf{u}_{-a})\big], \quad (5)$$

$$\bar r^{\pi_{-a}}(s, u_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi_{-a}(\cdot|s)}\big[r_a(s, u_a, \mathbf{u}_{-a})\big]. \quad (6)$$

The value that IQL implicitly chases is the optimal value of this perceived MDP, i.e. the *best response* of agent a to π_{-a}, defined as the fixed point of the best-response Bellman operator

$$(\mathcal{B}^{\mathrm{BR}}_{\pi_{-a}} Q)(s, u_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi_{-a}(\cdot|s),\, s'\sim\mathcal{T}}\Big[r_a(s, \mathbf{u}) + \gamma \max_{u'_a} Q(s', u'_a)\Big]. \quad (7)$$

We denote its fixed point $Q^{\mathrm{BR}}_a(s, u_a; \pi_{-a})$. For *fixed* π_{-a}, $\mathcal{B}^{\mathrm{BR}}_{\pi_{-a}}$ is a γ-contraction (Appendix B), so $Q^{\mathrm{BR}}_a(\cdot; \pi_{-a})$ is well defined.

This is a *best-response* value, *not* a joint-optimal or equilibrium value: it is optimal for *a only against the fixed* π_{-a}. During training the others keep learning, so the relevant target is $Q^{\mathrm{BR}}_a(\cdot; \pi^{\mathrm{cur}}_{-a})$ with π^{cur}_{-a} *moving*. Off-policy MARL thus rests on a *quasi-stationary* approximation: at each update we treat the current others' policies as fixed. Perceived non-stationarity is precisely the drift of this target, and it is what makes a fixed buffer inconsistent.

3.2 What exactly is mismatched

The buffer stores transitions $\tau_i = (s_i, u_{i,a}, \mathbf{u}_{i,-a}, r_i, s'_i)$ in which the others' actions were drawn from their *past* policy, $\mathbf{u}_{i,-a} \sim \pi^{\mathrm{past}}_{i,-a}(\cdot \mid s_i)$. The loss we would *like* to minimise draws them from the *current* policy:

$$\mathcal{L}^{\mathrm{cur}}_a(\phi_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\mathrm{cur}}_{-a}(\cdot|s)}\Big[\big(Q_{\phi_a}(s, u_a) - y_a\big)^2\Big], \quad (8)$$

$$\mathcal{L}^{\mathrm{past}}_a(\phi_a) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\mathrm{past}}_{-a}(\cdot|s)}\Big[\big(Q_{\phi_a}(s, u_a) - y_a\big)^2\Big], \quad (9)$$

where s and u_a are drawn from the buffer in both lines, while the target y_a, the reward r and the next state s' are *downstream* of $\mathbf{u}_{-a}$ (since $r = r_a(s, u_a, \mathbf{u}_{-a})$ and $s' \sim \mathcal{T}(\cdot \mid s, u_a, \mathbf{u}_{-a})$). Equations (8) and (9) thus differ only through the conditional law of $\mathbf{u}_{-a}$ given s; the environment's conditional law of (r, s') given $(s, u_a, \mathbf{u}_{-a})$ is policy-independent, which is exactly why a single change of measure on $\mathbf{u}_{-a}$ (below) suffices. In particular, the *state* distribution is, in both cases, that of the buffer: reweighting the others' actions does *not* turn the past state-visitation $d^{\mathrm{past}}(s)$ into the current one $d^{\mathrm{cur}}(s)$. Any action-level correction therefore fixes *which joint actions* are credited, not *which states* are visited—a distinction that matters under function approximation and that we make explicit rather than hide.

3.3 Importance reweighting, done correctly

Going from (9) to (8) is a change of measure. For any integrable f,

$$\mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\mathrm{cur}}_{-a}(\cdot|s)}[f] = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\mathrm{past}}_{-a}(\cdot|s)}[\rho\, f], \qquad \rho = \frac{\pi^{\mathrm{cur}}_{-a}(\mathbf{u}_{-a} \mid s)}{\pi^{\mathrm{past}}_{-a}(\mathbf{u}_{-a} \mid s)}, \quad (10)$$

provided the support condition $\operatorname{supp}\pi^{\mathrm{cur}}_{-a}(\cdot \mid s) \subseteq \operatorname{supp}\pi^{\mathrm{past}}_{-a}(\cdot \mid s)$ holds. With unmasked ϵ-greedy exploration every action keeps

positive probability, so ρ is finite; but when actions are *masked* (as in our Water-Bomber environment) the past support can miss actions that are likely now, and ρ diverges. This is not a numerical detail—it is where the method breaks.

Direct loss correction is unbiased but unstable. Multiplying each squared TD error by ρ_i yields an unbiased estimate of (8). Unfortunately ρ is a ratio of products of small probabilities over several agents; it is heavy-tailed, and a single large ρ_i produces a gradient spike. This is the empirically observed instability [3] that we revisit in Section 5.

Prioritisation is self-normalised, not unbiased. Instead of multiplying the loss, one can *sample* transition i with probability $P(i) = \rho_i / \sum_k \rho_k$ and drop the explicit factor. The resulting estimator of $\mathbb{E}_{\mathbf{u}_{-a} \sim \pi^{\text{cur}}_{-a}}[f]$ is the *self-normalised* importance estimator

$$\widehat{\mathbb{E}}_{\text{SNIS}} = \frac{\sum_i \rho_i f_i}{\sum_k \rho_k} \;\neq\; \frac{1}{N} \sum_i \rho_i f_i = \widehat{\mathbb{E}}_{\text{IS}}. \tag{11}$$

The two coincide asymptotically when $\mathbb{E}[\rho] = 1$, but the self-normalised one is *biased at finite batch size*. Crucially, the recommendation *not* to divide back by $P(i)$ (unlike prioritised experience replay [12]) is *not* a variance trick that preserves the objective: it is a *deliberate change of the optimised objective* towards the current-policy expectation. Stating this honestly removes the apparent contradiction between "we correct the expectation" and "we change the distribution without correction": we do the latter, on purpose.

Why the explicit ratio is not enough. Computing ρ_i requires (i) storing or recomputing $\pi^{\text{past}}_{i,-a}(\mathbf{u}_{i,-a} \mid s_i)$ and (ii) evaluating $\pi^{\text{cur}}_{-a}(\mathbf{u}_{i,-a} \mid s_i)$. This is feasible only for *queryable, discrete* policies with shared support. It fails for partially observable agents (whose policies condition on histories), continuous actions (densities, not masses), parametric or noisy exploration, and black-box or human teammates. These are exactly the realistic cases. We now replace the ratio by a quantity that survives all of them.

4 CONFORMAL REPLAY

The common object of this section is a *non-conformity score* that grades how incompatible a stored transition is with the other agents' current behaviour, and its calibration into distribution-free p-values. The score then drives both a *selection* rule (Section 4.3) and a *detection* rule (Section 4.4).

4.1 A non-conformity score for stored transitions

We seek a score $A(\tau)$ that is *large* when transition τ is stale, i.e. when its other-agent behaviour is unlikely under the present. We give three, in decreasing order of the access to the other agents they require.

(A) Policy-based scores. When the policies are queryable and the others explore with discrete, full-support actions, the *exact* score is the log importance ratio

$$A(\tau_i) = -\log \rho_i = -\log \pi^{\text{cur}}_{-a}(\mathbf{u}_{i,-a} \mid s_i) + \log \pi^{\text{past}}_{i,-a}(\mathbf{u}_{i,-a} \mid s_i) : \tag{12}$$

resampling transitions with weight ρ_i realises the self-normalised estimator (11), i.e. makes the replayed $\mathbf{u}_{-a}$ follow π^{cur}_{-a}, and recovers the importance-weighting of Foerster et al. [3]. It requires the *past* likelihoods $\pi^{\text{past}}_{i,-a}(\mathbf{u}_{i,-a} \mid s_i)$ to be stored with each transition.

When only the *current* policies can be queried, a cheaper, ratio-free surrogate is

$$A(\tau_i) = -\log \pi^{\text{cur}}_{-a}(\mathbf{u}_{i,-a} \mid s_i), \tag{13}$$

which scores high any stored joint action the others would now rarely take, needs no past likelihood, and has no support/blow-up problem. It is, however, a *coarser heuristic*: unlike $-\log \rho$ it does not target π^{cur}_{-a} exactly, and it over-penalises actions that are intrinsically rare yet still current (where $\pi^{\text{cur}}_{-a} \approx \pi^{\text{past}}_{-a}$ are both small, so $\rho \approx 1$). It is the fallback—together with the model-based score below—for the regimes (partial observability, continuous actions, black-box teammates) where ρ is unavailable.

(B) Model-based score. When the policies cannot be queried, learn a model $\hat{h}^{\text{cur}}(s', r \mid s, u_a)$ of the *currently perceived* dynamics from a sliding window of fresh transitions, and score a stored transition by its surprise,

$$A(\tau_i) = -\log \hat{h}^{\text{cur}}(s'_i, r_i \mid s_i, u_{i,a}), \tag{14}$$

or, for a mean predictor $\hat{m}^{\text{cur}}$ with residual covariance $\hat{\Sigma}$, by the Mahalanobis residual $A(\tau_i) = \|s'_i - \hat{m}^{\text{cur}}(s_i, u_{i,a})\|^2_{\hat{\Sigma}^{-1}}$. Because the others' policies enter the perceived kernel $P^{\pi^{\text{cur}}_{-a}}$, a change in π^{cur}_{-a} shows up as elevated prediction error: stale transitions are exactly the out-of-distribution ones. We deliberately do *not* assume the error has zero mean—a learned model is imperfect—and calibrate its nominal distribution instead, as follows.

4.2 From scores to conformal p-values

Maintain a calibration set $C = \{\tau^{\text{cal}}_j\}_{j=1}^n$ of *fresh* transitions, generated under the current policies and held out from fitting the score. Let $A^{\text{cal}}_j = A(\tau^{\text{cal}}_j)$. For a buffer transition τ_i with score $A_i = A(\tau_i)$, define the (inductive) conformal p-value

$$p_i = \frac{1 + \left|\{\, j : \; A^{\text{cal}}_j \geq A_i \,\}\right|}{n+1} \in (0, 1]. \tag{15}$$

A small p_i means τ_i is more non-conforming than almost all fresh transitions. Under the null hypothesis that τ_i is exchangeable with the calibration sample (its other-agent behaviour conforms to the present), p_i is (super-)uniform on $(0, 1]$ [20]. The construction is *distribution-free*: it inherits no Gaussian or zero-mean assumption from the score. When precision matters, a Mondrian / state-conditional variant restricts the calibration set to transitions whose (s, u_a) falls in the same region as τ_i, which guards against the model being merely less accurate in some part of the space.

4.3 Use 1: Conformal transition selection

We turn the p-values into a replay distribution. Following the large-batch recipe of LaBER [5], we draw a large batch, score it, and *subsample* a training mini-batch with a probability that *increases with conformity*, e.g. proportional to p_i, or a hard conformal filter $\mathbf{1}[p_i \geq \alpha]$ at level α (Algorithm 1). Intuitively we keep transitions that the present could plausibly have produced.

Algorithm 1 Conformal Likely-Transition Selection (per agent a)

Require: buffer $\mathcal{D}$, fresh calibration set C, large-batch size L, mini-batch size B, score $A(\cdot)$

while training **do**
  refresh C with recent on-policy transitions; compute $\{A_j^{\text{cal}}\}$
  sample a large batch $\{\tau_i\}_{i=1}^{L}$ from $\mathcal{D}$ (uniformly or with PER)
  compute scores $A_i = A(\tau_i)$ and p-values p_i via (15)
  set weights $w_i \propto p_i$ (or $w_i \propto \mathbf{1}[p_i \geq \alpha]$); subsample B indices $\sim w$
  compute up-to-date targets y_i and take an SGD step (3) on the mini-batch
end while

Algorithm 2 Conformal Drift Detection (per agent a)

Require: calibration set C, level δ, mixing over ε

$M \leftarrow 1$
for each newly observed transition τ_t **do**
  compute $A_t = A(\tau_t)$ and p_t via (15) against C
  $M \leftarrow M \cdot \int_0^1 \varepsilon\, p_t^{\varepsilon-1}\, \mathrm{d}\varepsilon$
  if $M \geq 1/\delta$ **then**
    declare drift: refresh C, down-weight/flush $\mathcal{D}$, reset $M \leftarrow 1$
  end if
end for

Relation to the importance ratio. With the exact score (12), conformal selection recovers a *rank-stabilised* form of the importance prioritisation of Foerster et al. [3]: a high ρ gives a low A, hence a high conformity p-value and a high selection weight. The difference is that the conformal calibration uses only the *rank* of A_i among fresh transitions, so the scale of ρ—and its heavy right tail (Section 3.3)—no longer destabilises training. The ratio-free score (13) and the model-based score extend the same rule to the cases where ρ cannot be computed.

What is and is not corrected. As in (11), this selection *deliberately* reshapes the training distribution towards the current perceived MDP; it is a calibrated heuristic reweighting, not an unbiased estimator, and—like every action-level scheme—it does not correct the state-visitation $d^{\text{past}}(s)$. We do not over-claim beyond this.

4.4 Use 2: Detecting drift of the perceived environment

Selection decides *which* transitions to trust; we also want to know *when* the buffer as a whole has gone stale, so as to refresh the calibration set, down-weight or partially flush the buffer, or widen exploration. This is a sequential change-detection problem on the stream of incoming p-values, for which conformal *test martingales* are the natural tool [9, 19, 21].

For a fixed $\varepsilon \in (0, 1)$ and p-values $p_1, p_2, \ldots$, define the power martingale, and its mixture,

$$M_t^{\varepsilon} = \prod_{j=1}^{t} \varepsilon\, p_j^{\varepsilon-1}, \qquad M_t = \int_0^1 \prod_{j=1}^{t} \varepsilon\, p_j^{\varepsilon-1}\, \mathrm{d}\varepsilon. \tag{16}$$

If, under the null H_0 (no drift), the p_j are conditionally super-uniform given the past, then $(M_t)_{t\geq 0}$ is a non-negative martingale with $M_0 = 1$ and $\mathbb{E}_{H_0}[M_t] = 1$ (Appendix D). Ville's inequality then bounds the *anytime* false-alarm probability,

$$\mathbb{P}_{H_0}\left(\sup_{t\geq 1} M_t \geq \tfrac{1}{\delta}\right) \leq \delta, \tag{17}$$

so raising an alarm when $M_t \geq 1/\delta$ controls the probability of *ever* falsely declaring drift by δ, with no need to fix a horizon (Algorithm 2).

Honest statement of guarantees. The bound (17) is *exact* only when the p_t are conditionally super-uniform under H_0. In RL this requires care: transitions are temporally dependent and the agent's own policy shapes the state distribution. We distinguish three regimes. *(i)* With an exact perceived model, or with inductive conformal calibration that preserves exchangeability [19], conditional validity holds and (17) is rigorous. *(ii)* With a held-out in-distribution calibration set we obtain marginal validity, which degrades if deployment visits regions under-represented in calibration. *(iii)* Otherwise the threshold is tuned on nominal trajectories for an acceptable empirical false-alarm rate. We report which regime we are in rather than claiming the strongest one by default.

4.5 Theoretical analysis: when selection starves, and how detection certifies it

We make two things precise: *(i)* why biasing replay towards currently-compatible transitions can move the learned fixed point *away* from the best response, and *(ii)* in what sense the test martingale of Section 4.4 *certifies* the selection rule of Section 4.3.

Setup. Fix π_{-a}^{cur}, so the perceived MDP $\mathcal{M}_a(\pi_{-a}^{\text{cur}})$ is stationary (Appendix B). Use linear features $Q_\theta(s, u_a) = \phi(s, u_a)^\top \theta$ with Φ of full column rank—taken *tabular or $\mathcal{S}_w$-local* for the cleanest statements below, since a general Φ couples in off-support next-state features through the bootstrap. A buffer measure μ over (s, u_a) is reweighted by conformal weights $w_i = g(p_i) \geq 0$ (soft $w_i \propto p_i$, or the hard filter $w_i = \mathbf{1}[p_i \geq \alpha]$); write the effective (s, u_a)-marginal $d_w \propto w \cdot \mu$, $D_w = \mathrm{diag}(d_w)$, and the D_w-weighted projection $\Pi_{D_w} = \Phi(\Phi^\top D_w \Phi)^{-1}\Phi^\top D_w$. We write the reweighted operator with the *current* kernel $P^{\pi_{-a}^{\text{cur}}}$ and reward $\bar{r}^{\pi_{-a}^{\text{cur}}}$; this presupposes the exact-ratio correction (Theorem 4.2(iii)). Under ratio-free scores (13) or hard filtering the replayed (s', r) instead follow π_{-a}^{past}, so the effective kernel is the *past* one—a second bias compounding the support effect below.

Proposition 4.1 (Reweighted fixed point). *Let $A_w := \Phi^\top D_w (I - \gamma P^{\pi_{-a}^{\text{cur}}})\Phi$ and $b_w = \Phi^\top D_w \bar{r}^{\pi_{-a}^{\text{cur}}}$. If A_w is positive definite, conformal-reweighted semi-gradient TD converges to the unique fixed point $\theta_w = A_w^{-1} b_w$, solving $\Phi\theta_w = \Pi_{D_w} \mathcal{B}^{\text{BR}}_{\pi_{-a}^{\text{cur}}}(\Phi\theta_w)$; if A_w is merely non-singular the fixed point exists and is unique but the iteration need not converge to it. A* sufficient *(not necessary [4]) condition for $\Pi_{D_w}\mathcal{B}^{\text{BR}}_{\pi_{-a}^{\text{cur}}}$ to be a γ-contraction in $\|\cdot\|_{D_w}$ is that d_w be stationary for $P^{\pi_{-a}^{\text{cur}}}$; conformal reweighting deliberately violates it, so contraction—and hence boundedness of the off-distribution fixed point [4, 17]—is not guaranteed, and restoring it requires an emphatic reweighting [14].*

Theorem 4.2 (Support starvation). *Let $\mathcal{S}_w = \mathrm{supp}(d_w) \subseteq \mathrm{supp}(\mu) = \mathcal{S}$, with Φ tabular or $\mathcal{S}_w$-local.* (i) *θ_w depends on $\mathcal{M}_a(\pi^{\mathrm{cur}}_{-a})$ only through the rows of A_w, b_w indexed by $\mathcal{S}_w$ (entries off $\mathcal{S}_w$ vanish).* (ii) *Hence $\Phi\theta_w$ solves the D_w-weighted projected Bellman equation supported on $\mathcal{S}_w$—the best response of the sub-MDP retaining $\mathcal{S}_w$; if an optimal best-response trajectory visits some $(s, u_a) \notin \mathcal{S}_w$, the greedy policy of θ_w can be arbitrarily suboptimal in $\mathcal{M}_a(\pi^{\mathrm{cur}}_{-a})$.* (iii) *The hard filter $w_i = \mathbf{1}[p_i \geq \alpha]$ deletes from $\mathcal{S}_w$ every stored (s, u_a) whose $\mathbf{u}_{i,-a}$ is improbable under π^{cur}_{-a}; when the reward-bearing optimal joint action is currently low-probability (exploration, or escaping a poor consensus) it is deleted and (ii) applies. Conversely, the exact score $A = -\log \rho$ with soft weights $w_i \propto \rho_i$ (no truncation) leaves the (s, u_a)-marginal unchanged—$d_w = \mu$, since $\mathbb{E}_{\pi^{\mathrm{past}}_{-a}}[\rho \mid s, u_a] = 1$, so $\mathcal{S}_w = \mathcal{S}$—and instead replaces the past perceived kernel by the current one inside the Bellman operator (the importance correction of Foerster et al. [3]), so θ_w targets $Q^{\mathrm{BR}}_a(\cdot; \pi^{\mathrm{cur}}_{-a})$.*

Proof: Appendix E. The theorem delimits the *safe regime* of conformal selection—soft, exact-ratio reweighting, which both preserves support and corrects the kernel, recovering 3. Outside it—ratio-free scores (13) or hard α-filtering—selection both starves support *and* leaves the kernel biased towards π^{past}_{-a}.

Remark (the Simultaneous-Attack collapse). In Simultaneous-Attack (single state; Section 5) the all-attack joint action is the only reward-bearing event. Once the iterate drifts towards all-abstain, π^{cur}_{-a} renders all-attack improbable and the hard filter zeroes its weight, so the rare exploratory transitions leave $\mathcal{S}_w$; by Theorem 4.2(iii) learning then locks onto the all-abstain consensus—predicting precisely the regime in which compatibility selection underperforms uniform replay. (With a single state the trajectory argument of (ii) degenerates; the operative mechanism is the weight-zeroing of the rewarding arm.)

Lemma 4.3 (Detection certifies selection). *Run the martingale of Section 4.4 on the* same *score A as the selector, with* fixed *(data-independent) weights $\tilde{w}_i$—as required by Barber et al. [2]; the data-dependent rule $w_i \propto p_i$ of Algorithm 1 is thus certified only in its frozen-weight form. Model the windowed p-values as i.i.d. with density q under drift (uniform under H_0)—an idealisation stronger than the conditional super-uniformity used elsewhere. Then the beyond-exchangeability coverage gap of the selector [2] obeys*

$$\Delta_{\mathrm{cov}} \;\leq\; \sum_i \tilde{w}_i \, d_{\mathrm{TV}}\big(\mathbb{P}^{\mathrm{cur}}_A, \mathbb{P}^{\mathrm{past}}_A\big) \;\leq\; \Big(\textstyle\sum_i \tilde{w}_i\Big)\sqrt{\tfrac{1}{2}\,\mathrm{KL}(q \,\|\, \mathrm{Unif})},$$

and the detector estimates the right-hand side: $\frac{1}{t}\log M_t \xrightarrow{a.s.} \mathbb{E}_q[\log g] = \mathrm{KL}(q \,\|\, \mathrm{Unif}) - \mathrm{KL}(q \,\|\, g) \leq \mathrm{KL}(q \,\|\, \mathrm{Unif})$, with equality for the log-optimal bet $g = q$.

Proof: Appendix F. A *log-optimal* detector ($g = q$) makes $\frac{1}{t}\log M_t \to \mathrm{KL}(q \,\|\, \mathrm{Unif})$, an asymptotically valid *upper* bound on the selector's miscoverage. The deployed power mixture (16) is not log-optimal, so its growth rate *under*-estimates $\mathrm{KL}(q \,\|\, \mathrm{Unif})$ by the regret $\mathrm{KL}(q \,\|\, g)$: it certifies selection only *qualitatively* (faster growth $\Rightarrow$ larger drift $\Rightarrow$ wider potential gap). The bound is numerically tight only when the bet is log-optimal *and* $A = -\log \rho$ (a sufficient statistic of $\pi^{\mathrm{past}}_{-a} \to \pi^{\mathrm{cur}}_{-a}$, so that score drift equals perceived-MDP drift); and the growth-rate estimate is asymptotic, whereas the alarm (17) is the anytime-valid object. In this precise sense, *detection certifies selection.*

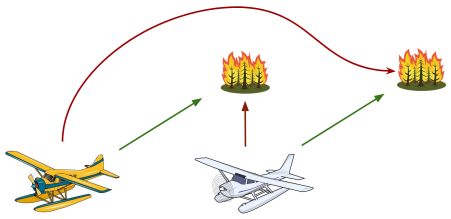


Figure 1: Water-Bomber. Two agents must extinguish all fires; the shared reward is granted only when every fire is covered, so the optimal joint behaviour (green) often requires one agent to take a longer route to free the other, rather than both rushing the nearest fire (red).

5 EXPERIMENTS

5.1 Two environments that isolate the phenomenon

Rather than benchmarking on SMAC [11], which entangles many factors, we use two minimal cooperative environments built so that episodic return is a clean measure of coordination.

Water-Bomber. A 5×3 grid with two water-bomber agents and fires appearing at random. An agent moves in four directions or fights a fire it stands on; *unavailable actions are masked.* The team receives a shared reward only once *all* fires are being extinguished, scaled so that an optimal policy reaches return 1. The environment requires genuine spatial coordination (Figure 1), and—because actions are masked—it is a setting where the importance ratio ρ can lose support (Section 3.3).

Simultaneous-Attack. A one-step, single-state bandit with N agents and two actions, *attack* or *abstain.* Attacking together succeeds; attacking without full participation is "suicidal". With $K = \sum_i a_i$ attackers we use the reward

$$r(\mathbf{a}) = \begin{cases} 1 & \text{if } K = N, \\ -\,r_{\mathrm{death}}\,K & \text{if } 0 < K < N, \\ 0 & \text{if } K = 0, \end{cases} \qquad r_{\mathrm{death}} = \frac{1}{N\,(2^{N-1} - 1)}. \tag{18}$$

This constant (corrected from the earlier $-1/(1 - 2^{-(N-1)})$, which had the wrong sign and scale) makes *abstention-vs-coordination* the whole difficulty: a uniformly random joint policy gets expected return 0, while the all-attack optimum gets 1 (proof in Appendix C). Coordination is exponentially rare under random play, so the buffer fills with "failed" transitions whose value depends entirely on whether teammates would still attack.

5.2 Protocol.

We use an IQL agent implemented from scratch (PyTorch, with TorchRL replay buffers), a dueling network for Water-Bomber, ϵ annealed from 0.99 to 0.05, and results averaged over multiple seeds (100 for Simultaneous-Attack, 10 for Water-Bomber). For each

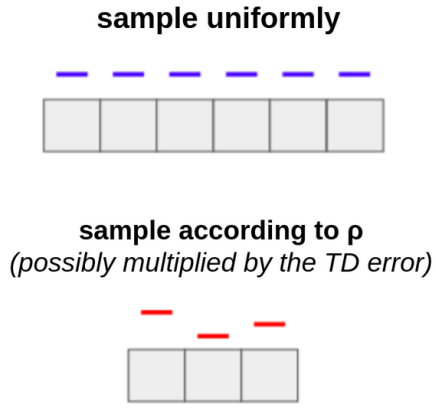


Figure 2: Large-batch-then-subsample replay. A large batch is drawn, each transition is scored, and a mini-batch is subsampled towards transitions compatible with the others' current policies. The importance ratio ρ is one possible score; conformal replay replaces it by a calibrated, ratio-free nonconformity score.



Figure 3: Motivation (Water-Bomber): increasing the replay-buffer size first improves then degrades performance, as stale transitions accumulate. The shaded band is the inter-seed spread.

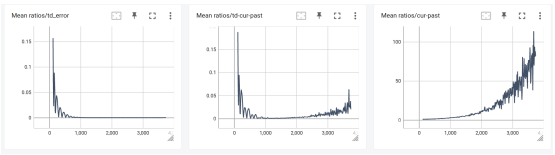


Figure 4: The importance ratio ρ is heavy-tailed and occasionally huge, which makes direct loss correction explode. Conformal calibration uses only the *rank* of the score and is immune to this.

method we report *average optimality* (normalised return) against environment steps.

5.3 Findings.

The problem is real (motivation). Growing the replay buffer (with the batch size scaled accordingly) helps *only up to a point* and then *hurts*, as old transitions become inconsistent with the others' current policies (Figure 3). A small buffer avoids the issue but forfeits sample efficiency—the dilemma this paper targets.

Selection by compatibility helps; raw ratio correction is unstable. On both environments, biasing replay towards transitions compatible with the others' current policies improves sample efficiency over uniform sampling and over TD-based prioritisation (PER, LaBER), while *directly* multiplying the loss by ρ destabilises training—the importance ratio spikes (Figure 4), confirming the analysis of Section 3.3. The explicit-ratio selection of [3] is the special case of Algorithm 1 with $A = -\log \rho$; replacing the ratio by the calibrated conformal score is what extends the method to masked actions and to settings where ρ is unavailable. The ratio-free conformal instantiation and the martingale-triggered buffer refresh (Algorithm 2) are the contributions under active evaluation; we report the protocol above so the comparison is reproducible.

6 RELATED WORK

The non-stationarity of the perceived environment is a long-standing theme in cooperative MARL [7]. Foerster et al. [3] stabilise experience replay by conditioning on a low-dimensional fingerprint of training time and by importance-weighting the loss; we revisit their ratio, correct its formulation, and remove its dependence on queryable discrete policies. Value-decomposition methods such as VDN [13] and QMIX [10], and the actor-critic MADDPG [6], address credit assignment but still inherit non-stationary individual learners; PPO-based on-policy methods [22] sidestep replay at a sample-efficiency cost. On the replay side, prioritised experience replay [12] and LaBER [5] reshape the sampling distribution by TD error or gradient norm; we borrow their large-batch machinery but prioritise by *policy conformity* rather than by error magnitude. Our detection component builds on conformal prediction [1, 20] and on conformal test martingales for change detection [19, 21], whose anytime validity follows from Ville's inequality [9, 18].

7 CONCLUSION AND FUTURE WORK

We recast the reuse of past experience in off-policy MARL as a question of *conformity to the present* and answered it with conformal prediction. A single non-conformity score, calibrated into distribution-free p-values, both selects replay transitions that match the other agents' current policies and, through a conformal test martingale, detects when the perceived environment has drifted—with an anytime false-alarm guarantee. The construction generalises importance-ratio reweighting to the realistic cases (masking, partial observability, continuous actions, black-box teammates) where the ratio is unavailable or unstable, and we were careful to state what it does and does not correct.

Several directions remain. The state-visitation mismatch is untouched and could be addressed with state-conditional (Mondrian) calibration or with corrections on $d(s)$. Validity in the dependent RL

stream deserves a dedicated analysis, possibly via e-processes [9]. Finally, the partially observable setting—where policies condition on histories and the model-based score (14) is most useful—is the natural next benchmark, together with value-decomposition backbones (VDN, QMIX) on SMAC.

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A CONVERGENCE OF Q-LEARNING, CORRECTED

The optimal Bellman operator (1) maps an *arbitrary* Q to a new function (it uses Q on the right-hand side, not Q^*). It is a γ-contraction in $\|\cdot\|_\infty$:

$$\begin{aligned}\left|(\mathcal{B}^*Q_1)(s,a) - (\mathcal{B}^*Q_2)(s,a)\right| &= \gamma\left|\mathbb{E}_{s'}\left[\max_{a'} Q_1(s',a') - \max_{a'} Q_2(s',a')\right]\right| \\ &\le \gamma\,\mathbb{E}_{s'}\left[\max_{a'}|Q_1(s',a') - Q_2(s',a')|\right] \\ &\le \gamma\,\|Q_1 - Q_2\|_\infty,\end{aligned}$$

using $|\max_x f(x) - \max_x g(x)| \le \max_x |f(x) - g(x)|$. Hence $\|\mathcal{B}^*Q_1 - \mathcal{B}^*Q_2\|_\infty \le \gamma\|Q_1 - Q_2\|_\infty$ with $\gamma < 1$; the factor γ is what makes it a contraction (a *non-expansion* would not suffice for Banach's theorem). The fixed point is Q^*.

B THE PERCEIVED BEST-RESPONSE OPERATOR

For fixed π_{-a}, the operator $\mathcal{B}^{\mathrm{BR}}_{\pi_{-a}}$ of (7) is a γ-contraction: the others' actions are averaged out into the perceived kernel $P^{\pi_{-a}}$ and reward $\bar{r}^{\pi_{-a}}$, reducing (7) to an ordinary optimal Bellman operator on $\mathcal{M}_a(\pi_{-a})$, to which Appendix A applies verbatim. Its fixed point $Q^{\mathrm{BR}}_a(\cdot;\pi_{-a})$ is the best response of a to π_{-a}. The MARL Bellman optimality equation

$$Q^{\mathrm{BR}}_a(s,u_a;\pi^{\mathrm{cur}}_{-a}) = \mathbb{E}_{\mathbf{u}_{-a}\sim\pi^{\mathrm{cur}}_{-a}}\left[r_a(s,\mathbf{u}) + \gamma\sum_{s'}\mathcal{T}(s' \mid s,\mathbf{u})\max_{u'_a} Q^{\mathrm{BR}}_a(s',u'_a;\pi^{\mathrm{cur}}_{-a})\right]$$

must be read as a best-response value *conditional on the current others' policies*, under the quasi-stationary approximation that they are held fixed during the update—not as a joint-optimal or equilibrium value.

C SIMULTANEOUS-ATTACK REWARD, CORRECTED

Let each agent attack independently with probability $\frac{1}{2}$ (uniform random policy), so $K = \sum_i a_i \sim \mathrm{Binomial}(N, \frac{1}{2})$. With the reward (18),

$$\mathbb{E}_U[r] = \underbrace{1\cdot\mathbb{P}(K=N)}_{=2^{-N}} - r_{\mathrm{death}}\sum_{k=1}^{N-1} k\,\mathbb{P}(K=k).$$

Since $\sum_{k=0}^{N} k\,\mathbb{P}(K=k) = \mathbb{E}[K] = N/2$ and the $k=N$ term is $N\cdot 2^{-N}$,

$$\sum_{k=1}^{N-1} k\,\mathbb{P}(K=k) = \frac{N}{2} - \frac{N}{2^N} = \frac{N}{2}\left(1 - 2^{-(N-1)}\right) = \frac{N\,(2^{N-1}-1)}{2^N}.$$

Therefore $\mathbb{E}_U[r] = 2^{-N} - r_{\mathrm{death}}\cdot N(2^{N-1}-1)\,2^{-N}$, which vanishes exactly for

$$r_{\mathrm{death}} = \frac{1}{N\,(2^{N-1}-1)} > 0,$$

giving a genuine *penalty* ($-r_{\text{death}}K < 0$) for partial attacks, $\mathbb{E}_U[r] = 0$ for random play, and $r = 1$ for the all-attack optimum. (For $N = 2$: $r_{\text{death}} = \frac{1}{2}$, and $\mathbb{E}_U[r] = \frac{1}{4} - \frac{1}{2} \cdot \frac{1}{2} = 0$.)

D CONFORMAL MARTINGALE VALIDITY

Let $\mathcal{F}_t = \sigma(p_1, \ldots, p_t)$. If, under H_0, each p_t is conditionally super-uniform, $\mathbb{P}(p_t \le u \mid \mathcal{F}_{t-1}) \le u$ for all $u \in [0,1]$, then for $\varepsilon \in (0,1)$,

$$\mathbb{E}\left[\varepsilon p_t^{\varepsilon-1} \mid \mathcal{F}_{t-1}\right] \le \int_0^1 \varepsilon u^{\varepsilon-1}\,\mathrm{d}u = 1,$$

so $M_t^\varepsilon = \prod_{j \le t} \varepsilon p_j^{\varepsilon-1}$ is a non-negative supermartingale with $M_0 = 1$ (a martingale when the p_t are exactly uniform). Mixing over ε preserves this. Ville's inequality [18] for non-negative supermartingales gives $\mathbb{P}_{H_0}(\sup_t M_t \ge 1/\delta) \le \mathbb{E}[M_0]\,\delta = \delta$, which is (17) and justifies the anytime threshold $M_t \ge 1/\delta$.

E PROOF OF THEOREM 4.2 (SUPPORT STARVATION)

Take Φ tabular or $\mathcal{S}_w$-local, so Π_{D_w} does not couple in features of states outside $\mathcal{S}_w$; for a general Φ the bootstrap term $\gamma\Phi^\top D_w P^{\pi^{\text{cur}}_{-a}}\Phi$ carries next-state features $\phi(l)$ with $l \notin \mathcal{S}_w$, and the clean "sub-MDP" reading of (ii) holds only when $\mathcal{S}_w$ is closed under $P^{\pi^{\text{cur}}_{-a}}$. *(i)* Both $A_w = \Phi^\top D_w(I - \gamma P^{\pi^{\text{cur}}_{-a}})\Phi$ and $b_w = \Phi^\top D_w \bar{r}^{\pi^{\text{cur}}_{-a}}$ carry $D_w = \mathrm{diag}(d_w)$ as a left factor; any (s,u_a) with $d_w(s,u_a) = 0$ contributes a zero row to D_w and so does not enter A_w or b_w. Hence θ_w is determined by $\{P^{\pi^{\text{cur}}_{-a}}, \bar{r}^{\pi^{\text{cur}}_{-a}}\}$ only through the rows indexed by $\mathcal{S}_w$. *(ii)* On $\mathcal{S}_w$ the normal equation $A_w\theta_w = b_w$ is the D_w-weighted projected Bellman equation of the sub-MDP that retains only state-actions in $\mathcal{S}_w$, whose solution is that sub-MDP's best-response value. Two perceived MDPs that agree on $\mathcal{S}_w$ but differ on $\mathcal{S} \setminus \mathcal{S}_w$ yield identical A_w, b_w, hence identical θ_w, while their true Q_a^{BR} differ; choosing the off-support values so that an omitted action is optimal makes $\arg\max_{u_a} \phi(s,u_a)^\top\theta_w$ suboptimal, with a gap as large as the reward range permits. *(iii)* For the hard filter $d_w(s,u_a) \propto \mu(s,u_a)\,\mathbf{1}[p \ge \alpha]$, and p is small exactly when $\mathbf{u}_{i,-a}$ is improbable under π^{cur}_{-a} (small $\pi^{\text{cur}}_{-a}(\mathbf{u}_{i,-a} \mid s)$, large score A); such pairs receive $d_w = 0$ and leave $\mathcal{S}_w$. For the converse, ρ is a function of $\mathbf{u}_{-a}$ (not of (s,u_a)) with $\mathbb{E}_{\pi^{\text{past}}_{-a}}[\rho \mid s,u_a] = 1$, so $w_i \propto \rho_i$ leaves the (s,u_a)-marginal unchanged ($d_w = \mu$, $\mathcal{S}_w = \mathcal{S}$) while, within each (s,u_a), re-weighting the stored $(\mathbf{u}_{-a}, s', r)$ from π^{past}_{-a} to π^{cur}_{-a}—replacing the past perceived kernel by the current one, on which the ordinary contraction of Appendix B applies and θ_w targets $Q_a^{\text{BR}}(\cdot;\pi^{\text{cur}}_{-a})$.

F PROOF OF LEMMA 4.3 (DETECTION CERTIFIES SELECTION)

Barber et al. [2], extending weighted conformal prediction [16], bound the coverage gap with *fixed* weights at the *residual* (score) level: coverage $\ge 1 - \alpha - \sum_i \tilde{w}_i\, d_{\text{TV}}\big(R(Z), R(Z^{(i)})\big)$, where R is the score map and $Z^{(i)}$ swaps calibration point i with the test point. With $R = A$ and the windowed i.i.d. model each swap perturbs a single coordinate, so $d_{\text{TV}}(R(Z), R(Z^{(i)})) = d_{\text{TV}}(\mathbb{P}_A^{\text{cur}}, \mathbb{P}_A^{\text{past}})$ and the bound collapses to the Lemma's first inequality. Let F_{past} be the (continuous, tie-free) past score CDF, so $p = 1 - F_{\text{past}}(A)$ is a monotone, invertible probability integral transform: under H_0, $p \sim \mathrm{Unif}[0,1]$; under drift its law has density q. As KL and total variation are invariant under invertible maps, $\mathrm{KL}(q \,\|\, \mathrm{Unif}) = \mathrm{KL}(\mathbb{P}_A^{\text{cur}} \,\|\, \mathbb{P}_A^{\text{past}})$ and likewise for d_{TV}; Pinsker's inequality $d_{\text{TV}} \le \sqrt{\frac{1}{2}\mathrm{KL}}$ then gives the second inequality. For the growth rate, $\frac{1}{t}\log M_t = \frac{1}{t}\sum_{j \le t} \log g(p_j) \to \mathbb{E}_q[\log g(p)]$ a.s. (SLLN), and $\mathbb{E}_q[\log g] = \mathrm{KL}(q \,\|\, \mathrm{Unif}) - \mathrm{KL}(q \,\|\, g) \le \mathrm{KL}(q \,\|\, \mathrm{Unif})$, with equality iff $g = q$. Hence only the log-optimal bet makes $\frac{1}{t}\log M_t \to \mathrm{KL}(q \,\|\, \mathrm{Unif})$ and turns the plug-in into an (asymptotic) *upper* bound on Δ_{cov}; the power mixture (16), equal to q only when q lies in the family $\{\varepsilon p^{\varepsilon-1}\}$, under-estimates it by the regret $\mathrm{KL}(q \,\|\, g)$ and certifies only qualitatively. The bound also requires fixed weights (the data-dependent selector $w_i \propto p_i$ falls outside the hypothesis), and equality of score drift with perceived-MDP drift requires A to be a sufficient statistic of $\pi^{\text{past}}_{-a} \to \pi^{\text{cur}}_{-a}$, i.e. $A = -\log\rho$. The estimate is asymptotic; the alarm (17) remains the anytime-valid object.